

ENDOCRINE PHYSIOLOGY OBJECTIVES

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ENDOCRINE PHYSIOLOGY OVERVIEW AND PRINCIPLES.

1. Know the major endocrine organs and the hormones they secrete.
2. What are the three major chemical classes of endocrine hormones?
3. Know the chemical class of all of the major hormones discussed.
4. Know the major steps in protein hormone and steroid hormone biosynthesis. What are the roles of the pre and pro segments of preprohormones?
5. Understand the mechanism of storage and secretion of protein/peptide hormones from endocrine cells.
6. Know the principle mechanisms used for inactivating or metabolizing hormones following secretion and the clinical significance of liver and renal disease on these processes.
7. Know predominant mechanisms used to metabolize steroid vs. protein hormones
8. Be able to distinguish the general mechanisms used for hormone blood transport between steroid hormones vs. protein hormones and catecholamines.
9. What are the physiological purposes of steroid hormone binding by plasma proteins? What is the functional difference between a bound vs. free (dissolved) hormone? What impact does predominant vs minimal hormone binding have on hormone concentrations in the blood?
10. Know the two basic classes of endocrine hormone receptors on target cells and the subtypes within each class.
11. Be able to identify the basic receptor type(s) utilized by the three principle structural classes of hormones.
12. Know the two basic types of second messenger pathways used by G-protein receptors?
13. Which receptor type(s) predominantly regulate target cells via a phosphorylation cascade?
14. Which receptor type(s) predominantly regulate target cells via a transcription/translation mechanism?
15. Why is steroid and thyroid hormone action slower on target cells compared to protein hormones?
16. Know the basic class of membrane receptor used by all major protein/peptide hormones. Be able to identify the three hormones discussed that act via a tyrosine kinase receptor instead of a G-protein receptor.
17. Understand the concepts of one hormone affecting many different processes; one hormone affecting many different aspects a single process and multiple hormones affecting a single process.
18. Know the principle factors that can alter normal hormone receptor action on target cells and the clinical relevance. What is the major factor that affects tissue hormone sensitivity? Be able to predict how changes in tissue sensitivity affect tissue responsiveness to hormones and understand the clinical relevance.

19. Be able to describe the basic mechanism of negative feedback inhibition control of hormone secretion.
20. Know the four basic physiological processes regulated by endocrine hormones and understand the complexities of a single hormone regulating multiple processes and single processes regulated by multiple hormones.
21. Understand that endocrine disorders invariably result from either deficient or excess hormone secretion and/or action and know the basic clinical algorithm used for clinical diagnosis.
22. Understand why the non-uniform secretion pattern of most hormones complicates the clinical diagnosis of endocrine disorders.

HYPOTHALAMIC & PITUITARY HORMONES.

1. Understand the anatomical and functional endocrine relationships between the hypothalamus and pituitary gland (hypothalamic-pituitary axis).
2. Know the different communication pathways between the hypothalamus and anterior pituitary versus posterior pituitary lobes.
3. Know all hormones produced and/or secreted from the hypothalamus and their basic properties (Actions, Target Tissue, Hormone Class, Synthesis, Secretion, Storage, Transport, Metabolism, Mechanism of Actions, Regulation).
4. Know all hormones produced and/or secreted from the anterior pituitary and their basic properties.
5. Understand the basic pathophysiology and clinical consequences of any of the hypothalamic/pituitary disorders discussed.
6. Understand the basic negative feedback mechanism that regulates secretion of peripheral hormone under the control of the hypothalamic-anterior pituitary axis.
7. Understand the basic classification scheme of hypothalamic-ant. pituitary disorders (primary, secondary, tertiary) and be able to deduce the classification of specific and general disorders given appropriate laboratory or functional data.
8. Be able to differentiate above classification and predicted lab results scheme between hormones regulated by hypothalamic-anterior pituitary axis (hormone regulates hormone) and those that are not (e.g. insulin).
9. Know the basic hormone properties of growth hormone and be able to distinguish from other anterior pituitary hormones as well as other protein hormones?
10. Describe growth hormone's specific actions on energy metabolism and compare and contrast with those of catecholamines, glucocorticoids, thyroid hormones, insulin and glucagon.
11. Explain the regulatory relationship between GH and insulin-like growth factor I (IGF-I)
12. Understand the pathophysiology of the clinical manifestations resulting from GH excess and GH deficiency and be able classify as primary or secondary based on etiology, patient symptoms, and lab findings.
13. Be able to explain how pituitary diabetes develops in relation to GH metabolic actions.

14. Be able to describe the hypothalamic-pituitary feedback mechanisms that regulate GH secretion and know the major physiological factors that stimulate or inhibit GH secretion.
15. Understand how sleep influences GH secretion and how it differs from other sleep-regulated and circadian affected hormones such as prolactin and cortisol.
16. Be able to describe the general lifetime secretion pattern of growth hormone.
17. Know the basic hormone properties of prolactin and be able to compare and contrast with other anterior pituitary hormones as well as other protein hormones.
18. Describe the hypothalamic-pituitary feedback mechanisms that regulate prolactin secretion and know the major physiological factors that stimulate or inhibit GH secretion.
19. Understand how hypothalamic hormone regulation of prolactin differs from other anterior lobe hormones and be able to explain why interruption of the hypothalamic-pituitary portal circulation increases prolactin secretion relative to other anterior pituitary hormones
20. Explain open loop regulation in relation to prolactin regulation.
21. Understand how sleep influences prolactin secretion and how it differs from other sleep-regulated and circadian affected hormones presented later.
22. Know the major physiological actions of prolactin including the distinct role of prolactin during lactation relative to oxytocin.
23. Be able to predict the major pathophysiological manifestations expected during hyperprolactinemia including reproductive impairments in males as well as females.
24. Be able classify prolactin disorders as primary or secondary based on etiology, patient symptoms, and lab findings.
25. Know the basic hormone properties of the two posterior pituitary hormones and compare and contrast with anterior pituitary hormones as well as other protein hormones.
26. Describe the unique mechanism of posterior pituitary hormone biosynthesis and storage.
27. Know the physiological actions of ADH and oxytocin and the principle physiological regulators ADH and oxytocin secretion.
28. Know the feedback loops that regulate secretion of posterior pituitary hormones including positive feedback & open-loop regulation of oxytocin secretion.
29. Be able to describe the "Anticipatory Reflex" related to oxytocin's actions in milk ejection.
30. Be able to predict the relative effects of changes in plasma osmolality on ADH secretion, renal H₂O reabsorption, and urine osmolality.
31. Describe the pressor effect of vasopressin and be able to recognized clinical circumstances when ADH may exert this action.
32. Be able to predict the pathophysiological manifestation of ADH disorders based on functional and clinical laboratory information.

THYROID HORMONES.

1. Know the unique hormone properties of thyroid hormones relative to other amino-acid derived hormone as well as protein and steroid hormones.
2. Know the specific cellular transport mechanisms used for iodide uptake into thyroid follicular cells from plasma and out into the colloid.
3. Be able to describe the specific steps in thyroid hormone biosynthesis from tyrosine residues including the role of thyroglobulin and specific actions of thyroidal peroxidase (TPO).
4. List and distinguish the major biosynthesized thyroid hormone products and the major thyroid gland product secreted into the blood
5. Know that the half-life of T4 is longer than T3, but that T3 is more potent than T4 and understand why.
6. Be able to explain the general routes of metabolism of T3 & T3 after secretion.
7. Explain why clinical manifestations related to impaired thyroid hormone biosynthesis are typically delayed for several weeks to months after the primary impairment develops
8. Distinguish the mechanism of action of thyroid hormones on target cells from other amino acid derived hormones and protein and steroid hormones.
9. Be able to describe the major physiological actions of thyroid hormones including the effect of thyroid hormones on metabolic rate and calorogenesis.
10. Be able to describe thyroid hormone actions on protein, carbohydrate, & lipid metabolism
11. Compare and contrast the metabolic actions of thyroid hormones with those of growth hormone, catecholamines, glucocorticoids, insulin and glucagon.
12. Be able to predict how metabolic actions are altered by hypothyroidism or hyperthyroidism.
13. Know effects of thyroid hormones on growth & development of major organ systems and be able to predict pathophysiological manifestations during hypothyroidism or hyperthyroidism.
14. Be able to describe the hypothalamic-anterior pituitary feedback mechanism that regulates thyroid hormone secretion including the the basic hormone properties of TRH and TSH.
15. Be able to classify primary and secondary hyperthyroid and hypothyroid conditions based on etiology, patient symptoms, and lab findings including Graves' disease, myxedema, and neonatal hypothyroidism (Cretinism).

PARATHYROID HORMONE AND VITAMIN D: Hormonal Regulation of Ca^{++} & PO_4 .

1. Know the general distribution & functions of calcium in the body.
2. Know the compartment of body calcium directly "sensed" by calcium regulating hormones
3. Know the major hormones that regulate calcium & phosphate homeostasis and their principle target tissues.
4. Know the basic hormone properties of parathyroid hormone and compare and contrast with other endocrine hormones.

5. Know the specific actions of PTH in the kidneys including the specific sites of action along the renal tubules related to regulation of calcium and phosphate homeostasis.
6. Understand why PTH has only limited capacity to reabsorb calcium from renal tubules.
7. Rationalize the physiological benefit of PTH's action of increasing PO₄ excretion in relation to calcium homeostasis.
8. Know the specific role of PTH in the regulation of vitamin D biosynthesis.
9. Review the basic structure and histology of cortical bone as it relates to the calcium & phosphate regulatory action of hormones.
10. Know what hydroxyapatite is and its relationship with soluble bone salts and bone fluid.
11. Explain bone remodeling in terms of bone synthesis and resorption and in relation to the roles of osteoclasts, osteoblasts, and osteocytes.
12. Know the major factors that regulate PTH secretion.
13. Describe the events that occur during the rapid and slow phases of PTH actions in bone in response to acute and chronic hypocalcemia.
14. Know the pathophysiological manifestations resulting from primary hyperparathyroidism and hypoparathyroidism and relate to the physiological actions of PTH.
15. Be able to predict laboratory diagnostic parameters associated with hyperparathyroidism and hypoparathyroidism
16. Know the basic hormone properties of Vitamin D and compare and contrast with other steroid hormone as well as protein hormones.
17. Explain the biosynthesis of 1,25(OH)₂cholecalciferol from skin to the liver to kidneys.
18. Know the physiological actions of vitamin D that result in increased calcium and phosphate absorption from the intestines
19. Know the factors that regulate vitamin D biosynthesis.
20. Know the general clinical manifestations that occur as a result of rickets or osteomalacia and be able to relate these symptoms to the normal physiological actions of vitamin D
21. Know the basic hormone properties of calcitonin
22. What are the basic actions of calcitonin in relation to calcium regulation?
23. Be able to describe the dynamics and sequence of events during the integrated hormonal response (PTH, Vitamin D & calcitonin) to hypocalcemia and hypercalcemia.

ADRENAL CORTEX HORMONES.

1. Know the structural class of adrenal cortex hormones, the general class of hormone produced by each histological zone, and the predominant hormone secreted by each zone.
2. Know the basic properties of glucocorticoids and compare and contrast with other adrenal cortical hormones as well as protein and amino-acid derived hormones..
3. Be able to describe the physiological actions of glucocorticoids and its target tissues including actions on energy stores of muscle, fat and liver.
4. Know the net effects of epinephrine actions on plasma glucose, FFA, and amino acids.
5. Compare and contrast the metabolic actions of glucocorticoids with those of growth hormone, catecholamines, thyroid hormones, insulin and glucagon.
6. Describe the general processes related to inflammation that are affected by glucocorticoids that account for their anti-inflammatory effect?
7. Understand the role of glucocorticoids in the stress response and how it is distinguished from the fight or flight response mediated by catecholamines.
8. Define Cushing's disease and be able to classify this condition and other hyperglucocorticoid disorders based on clinical laboratory measurements of trophic and peripheral hormones.
9. Correlate clinical manifestations associated with Cushing's disease/syndrome with the normal physiological actions of glucocorticoids including development of adrenal diabetes.
10. Define Addison's disease and be able to classify this condition and other hypoglucocorticoid disorders based on clinical laboratory measurement of trophic and peripheral hormones.
11. Correlate clinical manifestations associated with Addison's disease with the normal physiological actions of glucocorticoids.
12. Know the basic properties of mineralcorticoids and compare and contrast with other adrenal cortical hormones as well as protein and amino-acid derived hormones.
13. Know the physiological actions of mineralcorticoids and its target tissues including the renal tubules.
14. Know the major regulatory mechanism that controls aldosterone secretion including the roles of renin and angiotensin II.
15. Know the pathophysiological manifestations developed from a mineralcorticoid deficiency and excess and be able to correlate with the normal physiological actions of mineralcorticoids.
16. Become familiar with interpreting the biochemical schematic outlining the biochemical biosynthesis of steroid hormones from cholesterol.
17. Understand why only certain steroids are produced in particular zones of the adrenal cortex.
18. Understand the overlapping functional nature of many adrenal cortex steroids and the clinical implications of hyper- or hyposecretion.

19. Understand conceptually how deficient secretion of one P450 enzyme results in excess secretion of upstream intermediates and deficiency of hormone synthesis downstream from its action
20. Be able to predict the steroid hormones that would be deficient and those would be in excess as a result of a given P450 enzyme deficiency (e.g. 21-hydroxylase deficiency)

ADRENAL MEDULLA HORMONES.

1. Know that catecholamines are biosynthesized for tyrosine by the adrenal medulla and that the major secretion product is epinephrine.
 2. Understand why a complete adrenalectomy would decrease the epinephrine to norepinephrine ratio in blood.
 3. Know the basic hormone properties of catecholamines and be able to compare and contrast to other amino acid derived hormones and to protein and steroid hormones.
 4. Be able to describe the physiological actions of adrenal catecholamines on energy metabolism including actions on muscle, fat and liver energy stores.
 5. Know the net effects of epinephrine actions on plasma glucose, FFA, lactate, glycerol, and ketones.
 6. Compare and contrast the metabolic actions of adrenal catecholamines with those of growth hormone, glucocorticoids, thyroid hormones, insulin and glucagon.
 7. What are the general manifestations related to adrenal medulla insufficiency, sympathetic autonomic insufficiency? Medullary hypersecretion?
 8. Be able to correlate pathophysiological manifestations to the normal physiological actions of catecholamines.
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